

Rohit Korpai

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Summary

Computer Science (AI & ML) undergraduate and Software Engineer with hands-on experience building scalable, full-stack and AI-integrated systems. Led end-to-end development of a large-scale data automation pipeline as Machine Learning Intern at TC Global, processing data across hundreds of distributed web sources. Gained applied experience in data analysis, statistical techniques, and visualization through a Data Analyst internship at Samatrix Consulting. Strong foundation in data structures, algorithms, distributed systems concepts, and software engineering principles, with proven ability to design, develop, and deploy production-grade solutions independently and collaboratively. Built multiple AI-driven, real-time applications recognized in national hackathons, including 2nd Runner-Up at the Microsoft Netwebatron AI for Good Hackathon. Passionate about applying computer science fundamentals to solve real-world, large-scale engineering challenges. Actively interested in research, with a focus on applying ML and AI to real-world systems.

Professional Experience

Machine Learning Intern

Feb 2025 – Jul 2025

TC Global (The Chopras)

New Delhi, India

- Designed and led the end-to-end development of a scalable, universal web crawler infrastructure using **Python**, **BeautifulSoup**, **Regular Expressions** and **Selenium** to automate structured data extraction from hundreds of distributed global university websites.
- Engineered **ML-powered data classification pipelines** to automatically categorize and validate extracted content — including program details, tuition fees, and application deadlines — reducing noise and improving structured data quality at scale.
- Applied **Natural Language Processing (NLP)** techniques for intelligent parsing of unstructured web content, enabling consistent extraction across diverse and dynamic site layouts.
- Built and fine-tuned **anomaly detection models** to identify inconsistent or corrupt data entries within the pipeline, ensuring downstream data integrity and reliability.
- Collaborated cross-functionally with engineering and business teams to integrate ML-enhanced real-time updates into internal databases, optimizing pipeline performance and decision-making speed.
- Delivered a production-grade ML-integrated system that significantly reduced manual effort and turnaround time, directly improving organizational efficiency across student advisory operations.

Data Analyst Intern

Jun 2025 – Jul 2025

Samatrix Consulting Pvt. Ltd.

India (Hybrid)

- Collected, cleaned, and processed large datasets from multiple sources, applying statistical techniques to ensure data quality and consistency.
- Performed exploratory data analysis to identify trends, patterns, and anomalies, translating business requirements into structured analytical tasks.
- Developed and implemented interactive data visualization dashboards to effectively communicate findings and support data-driven decision-making.
- Documented data analysis processes and methodologies; presented findings and progress updates in team meetings.

Web Development Intern

Jun 2024 – Aug 2024

Pine Air Pvt. Ltd.

New Delhi, India

- Developed and debugged front-end features using **HTML**, **CSS**, and **JavaScript**, improving site stability and reducing reported UI defects.
- Maintained and enhanced a production company website, improving user experience, interface performance, and cross-browser compatibility.
- Implemented **on-page SEO** best practices including semantic HTML structuring, meta tag optimization, and page speed improvements, contributing to better search engine visibility and organic reach.

Skills

Languages: Python, Java, C, R, JavaScript, SQL

Web & Full-Stack: HTML, CSS, JavaScript, REST APIs, Front-End & Back-End Development

AI / ML: Machine Learning, Deep Learning, Large Language Models (LLM Integration), Computer Vision, VGG16

Databases & Data: MySQL, MongoDB, Data Pipelines, Data Automation, Web Scraping (Selenium, BeautifulSoup)

Data Analytics: Power BI, Tableau, Excel, NumPy, SciPy, Matplotlib, Seaborn

Cloud & Tools: AWS (Elastic Beanstalk), Git, GitHub, VS Code, Streamlit

Core CS: Data Structures & Algorithms, Object-Oriented Design, Distributed Systems Concepts, System Design Fundamentals

Projects

Optiflow: Smart AI Traffic & Accident Detection System | [source code](#)

Apr 2026

- Engineered a real-time AI-driven traffic management system to analyze live traffic data, optimize flow, and enhance road safety at scale.
- Designed intelligent detection algorithms to identify congestion patterns, accident-prone zones, and automatically trigger real-time alerts to authorities.
- Implemented data-driven decision logic for proactive route optimization and road-safety interventions; secured **2nd Runner-Up** at the Microsoft Netwebatron AI for Good Hackathon.

WardSense: AI-Driven Pollution Action & Accountability System | [source code](#)

Apr 2026

- Built a full-stack citizen-government platform enabling real-time reporting of pollution sources and ward-wise Air Quality Index (AQI) monitoring.
- Designed a government-facing dashboard for issuing public alerts, tracking inter-departmental actions, and monitoring pollution-control progress across regions.
- Integrated an LLM-based analytics module to generate AI-driven insights on pollution causes, affected population, AQI trends, and pollutant concentrations (NO₂, CO, PM).

MatchMed: Machine Learning-Based Organ Matching System | [source code](#)

May 2024

- Developed an ML-based matching engine to assist healthcare professionals with organ-donor compatibility decisions; [source code](#).
- Designed a compatibility-scoring algorithm displaying overall and factor-level match percentages between recipients and donors.
- Integrated **MongoDB** for donor data storage and built a front-end interface for hospital staff to manage recipient matching workflows.

Pneumonia Detection Using Deep Learning | [source code](#)

May 2024

- Sourced chest X-ray data from the **National Health Authority of India** and fine-tuned a **VGG16** pretrained CNN model to accurately classify the presence or absence of pneumonia in lung X-rays, demonstrating real-world applicability in medical imaging.

Education

Bachelor of Technology, Computer Science Engineering (AI & ML)

Shree Guru Gobind Singh Tricentenary University

Expected 2027

Haryana, India

- SGPA (Semester V): 9.42/10.0

Certifications

- **AWS APAC Solutions Architecture Virtual Experience Program** – Forage (Dec 2025): Designed a scalable hosting architecture using AWS Elastic Beanstalk for a high-growth client.
- **Deloitte Australia Data Analytics Job Simulation** – Forage (Dec 2025): Built interactive Tableau dashboards and conducted data analysis for business insights.
- **Database Management Systems (DBMS)** – IIT Kharagpur (2024): Database design, normalization, and SQL query development.
- **Artificial Intelligence & Machine Learning** – IIT Delhi (2024): Applied AI/ML concepts to real-world problem-solving.
- **Full Stack Development** – SGT University (2024): Built web applications using HTML, CSS, JavaScript, and SQL.

Achievements, Leadership & Research Interests

- **2nd Runner-Up** – Microsoft Netwebatron AI for Good Hackathon, Gurugram (2026), among 10+ national-level hackathons in AI, Healthcare Technology, Sustainability, and Smart Cities.
- **3rd Prize** – Innoverse'36 Hackathon, SGT University.
- **Member**, Atal Community Innovation Center, SGT University (Oct 2024 – Present): Contributing to technology-driven projects and research focused on social impact and sustainable development.